

Anatomy of an AI-Assisted Investigation

A Methodological Case Study: Deconstructing the DESI DR2
Evolving-Dark-Energy Signal Through Iterative Human–AI Collaboration

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Abstract

This document is a methodological case study, not a physics result. It records *how* an independent researcher, working in sustained collaboration with a large language model (LLM), conducted a multi-stage investigation into whether the DESI DR2 preference for evolving dark energy reflects new physics or an accumulation of measurement effects. The underlying scientific conclusion — that the signal is a stack of individually weak, largely mundane contributions — is summarized only as context; the focus here is the *process*. We document the chronological research trajectory and its pivot points (§2); compare the AI-assisted workflow against conventional human research practice and extract transferable lessons (§3); state limitations, ethical considerations, and a formal reproducibility protocol (§4); and, most substantially, red-team the methodology itself against AI-specific failure modes — confirmation-bias feedback loops, hallucination, alignment-style sycophancy drift, and pipeline integrity (§5). The central methodological finding is that the value of the AI collaboration lay less in generating answers than in *industrializing self-refutation*: the workflow’s defining feature was the repeated, cheap construction of controls, null simulations, and bug-checks that overturned the collaboration’s own intermediate claims. We argue this is the property human researchers should selectively adopt, and the one most vulnerable to the failure modes we enumerate.

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1 Introduction and scope

The scientific object of the investigation was a live controversy: the Dark Energy Spectroscopic Instrument (DESI) Data Release 2 reported a $3\text{--}4\sigma$ preference for a time-evolving dark-energy equation of state (an apparent “phantom crossing,” $w_0 > -1$, $w_a < 0$) over a cosmological constant [1]. Whether this reflects new physics or systematics has divided the field [2, 3, 6].

This paper does not adjudicate that question. It documents the *methodology* of one independent attempt to pressure-test the signal, conducted as an extended dialogue between a human researcher and an LLM-based assistant with code-execution, web-search, and persistent-memory tools. We adopt a five-perspective structure — historical, comparative, ethical, adversarial, and architectural — because the honest evaluation of an AI-assisted method requires precisely these distinct lenses. We write in a single voice throughout: the “panel” is an organizing device for scope, not a claim that five independent authorities contributed.

Scope disclaimer

The physics conclusions referenced here are provisional, method-conditional, and reproduced in a deliberately simplified analysis pipeline (compressed CMB priors, magnitude proxies) that inflates absolute significances. They are used only to illustrate the *process*. No claim in this document should be read as a validated cosmological result; see §4.

2 The research trajectory: a chronological methodology

(*Perspective 1: research methodologist & historian.*)

The investigation did not proceed linearly from hypothesis to test. It proceeded as a *funnel with reversals* — broad theoretical framing, progressive narrowing to a tractable empirical target, and several sharp reversals where a self-imposed check overturned a claim the collaboration had already adopted. The reversals, not the confirmations, were the load-bearing events.

2.1 Phase I — Theoretical framing and elimination

The starting point was the broadest possible framing: the cosmological-constant problem and the space of dark-energy explanations. An early decision — consequential in retrospect — was to *not* build a new dark-energy model, after a literature scan established that the holographic/quintom/interacting-sector model-building space is saturated [7]. This redirected effort from theory generation toward measurement critique, where an independent researcher with a validated statistical toolkit could contribute without competing against large collaborations. Formal work in this phase (a growth-channel

classification of geometric dark energy) established a reusable result: event-horizon holographic dark energy is provably smooth sub-horizon, closing its structure-growth observability.

2.2 Phase II — Narrowing to the supernova anchor

The pivot to empiricism came from a specific foundational paper: the a_B intercept diagnostic of Huang et al. [2], which is *independent of the fiducial matter density Ω_m* . This mattered because the entire difficulty of the field is an Ω_m -degeneracy — a magnitude-only probe cannot separate a real feature from an Ω_m shift. The a_B construction sidesteps it. The methodology here was *replication-first*: rather than accept the paper’s claim, the collaboration re-fetched the public DESY5 and Pantheon+ data and rebuilt the diagnostic from scratch.

Pivot 1 — the survey-lineage split

A first, naive redshift cut ($z < 0.1$) reproduced the discrepancy only weakly (1.0σ). Reframing the split by *survey lineage* (external low- z anchor vs. DES-native photometry, using the IDSURVEY field) raised it to 7.4σ , with a Pantheon+ control at 0.1σ . The lesson: the systematic lived in the *seams between heterogeneously calibrated datasets*, not in redshift. Choosing the physically correct partition, not a convenient one, was the whole game.

2.3 Phase III — Loop closure and its first correction

The collaboration then “closed the loop”: injected the measured offset into a w_0w_a fit against BAO and CMB to test whether correcting it collapsed the phantom-crossing preference. It did, with a monotonic dose–response and correct sign dependence. But this produced the investigation’s **first major reversal**: the simplified pipeline reported that correction removed 73–92% of the signal, whereas the peer-reviewed DES-Dovekie recalibration [4, 5] showed the rigorous number is $\sim 1\sigma$ of $\sim 4\sigma$ ($4.2\sigma \rightarrow 3.2\sigma$). The collaboration’s own intermediate claim was an overstatement, caught only by cross-checking against the literature. The overclaim was retracted in the working record.

2.4 Phase IV — The residual and a self-caught bug

Attention turned to the surviving $\sim 3.2\sigma$. A residual-shape analysis initially suggested a redshift-dependent feature (a candidate progenitor-age-bias signature). This was the **second reversal**: on re-examination, the apparent z -shape was an artifact of *unweighted* binning dominated by a handful of very-large-error supernovae. Error-weighting — what the likelihood actually does — flattened it. The age-bias template returned 0.1σ ; further, age-bias and evolving- w templates were shown to be 87% degenerate in shape, i.e. intrinsically indistinguishable from $\mu(z)$ alone.

2.5 Phase V — Migrating to the BAO sector and the decisive coordinate change

Recognizing the supernova side was near its public-data ceiling, the investigation migrated to the BAO tracers. A leave-one-out test found that the presumed “LRG1 anomaly” was only 1.6σ ; the real driver was LRG2 (3.0σ raw, $\sim 2.4\sigma$ after look-elsewhere). The **third and most instructive reversal** came from an analogy proposed by the human collaborator — whether a Smith-chart-like representation could help. The literal tool did not map, but the underlying instinct (“rotate into coordinates where physics becomes direction”) led to the Alcock–Paczynski decomposition (D_V/r_d isotropic vs. $F_{AP} = D_M/D_H$ anisotropic). In the $(D_M/r_d, D_H/r_d)$ basis the two tracers looked like one coherent line-of-sight pull; in the physically natural AP basis they *dissolved* into two unrelated

Phase	Action	Reversal / lesson
I	Theoretical framing; declined to build a new model	Model space saturated $\Rightarrow$ redirect to measurement critique
II	Rebuild a_B diagnostic on public data	Survey-lineage split (7.4σ) $\gg$ naive redshift cut (1.0σ)
III	Loop-closure fit	Reversal 1: “removes the bulk” overstated; true effect $\sim 1\sigma$
IV	Residual-shape / age-bias test	Reversal 2: z -shape was an unweighted-binning artifact
V	BAO leave-one-out + AP rotation	Reversal 3: coordinate change dissolved apparent coherence
VI	Pre-registered DR3 test; ledger	Question is data-limited; commit falsifiable prediction

Table 1: The research trajectory as a funnel with reversals. Each reversal was produced by a self-imposed check overturning a claim the collaboration had already adopted.

discrepancies (one isotropic, one anisotropic) — a more deflationary, fluctuation-favoring reading. A BOSS DR12 cross-check added mild independent evidence for a DESI-specific effect.

2.6 Phase VI — Pre-registration and consolidation

The terminal recognition was that the question is *data-limited*: no further analysis of current public data can separate a residual systematic from real physics. The methodological response was to construct a *pre-registered* decision test for DR3 — a quantitative χ^2 boundary, with a systematic-error-floor correction, committed to before the data arrive. The full trajectory was consolidated into a persistent, tiered ledger (SUPPORTED / OPEN / CONJECTURAL).

3 Comparative analysis: AI-assisted vs. conventional practice

(*Perspective 2: AI subject-matter expert & cognitive scientist.*)

The central comparative claim is narrow and defensible: the AI collaboration did not outperform a human expert at *judgment*, but it radically lowered the cost of *self-refutation*, and that cost structure changed the research dynamics.

3.1 Where the asymmetry lies

In conventional practice, constructing a control, a null simulation, or an independent cross-check is expensive — each is hours of coding. That cost imposes a rational bias: researchers check the claims they most doubt, and let plausible-looking intermediate results stand. In the AI-assisted workflow, a control or a 200-realization null Monte Carlo was minutes of generated-and-executed code. This inverted the economics: it became cheaper to *try to break* a result than to accept it. Every one of the three reversals in §2 was a direct consequence of this inversion — checks that a cost-rational human might have skipped were run by default.

3.2 Transferable practices for human researchers

These are actionable independent of whether one uses an LLM:

Dimension	Conventional practice	AI-assisted workflow (observed)
Cost of a control/null test	High (hours) $\Rightarrow$ selective	Low (minutes) $\Rightarrow$ default
Literature reconciliation	Periodic, batched	Continuous, per-claim
Breadth of methods tried	Bounded by fixed effort	Wide (many quick probes)
Depth / rigor per method	High	Variable; pipeline-simplified
Failure mode	Under-checking	Over-confident fluency
Dominant risk	Missed systematic	Plausible fabrication

Table 2: Comparative cost structure. The AI workflow’s advantage (cheap self-refutation) and its dominant risk (fluent overconfidence) are two faces of the same property.

1. **Budget self-refutation as a first-class activity.** Pre-commit to a control and a null simulation for every headline claim, before believing it. The AI made this cheap; humans can make it *mandatory*.
2. **Treat coordinate/parametrization choice as a hypothesis.** The AP-basis reversal shows a conclusion can be an artifact of representation. Re-derive key results in a second, physically motivated basis before publishing.
3. **Weight before you look.** The age-bias artifact was an unweighted-mean error. Inspect residuals in the metric the likelihood uses, not the eye’s.
4. **Maintain a tiered, persistent ledger.** Separating SUPPORTED / OPEN / CONJECTURAL claims, with explicit method-conditionality flags, prevents compression drift — the slow hardening of a hedged result into an asserted one across a long project.
5. **Pre-register the falsifier.** When a question is data-limited, the honest deliverable is a quantitative prediction committed to before the arbitrating data arrive.

The counter-lesson. Speed is not rigor. Every quantitative result the AI produced required an independent bug-check, and several absolute numbers remained pipeline-conditional throughout. The workflow’s throughput is only an asset when paired with the adversarial discipline of §5; without it, the same speed manufactures confident errors faster.

4 Limitations, broader impacts, and reproducibility

(Perspective 3: ethics & reproducibility auditor.)

4.1 Methodological limitations

- **Pipeline simplification.** Absolute significances rest on compressed CMB distance priors and magnitude proxies, which inflate values relative to a full likelihood. The robust content is in *directions*, *relative* shifts, and *reproduced structure*; several absolute numbers (notably the LRG2 isotropic significance) are explicitly method-conditional and flagged as requiring a full uncompressed-likelihood recomputation.
- **Non-independence of cross-checks.** The BOSS/DESI comparison partially overlaps in sky and galaxies, weakening its status as an independent probe.

- **Single-operator, single-model.** All AI assistance came from one model family in one extended session context; results may carry model-specific idiosyncrasies not audited here.

4.2 Broader impacts and ethical considerations

- **Automation of premature certainty.** A workflow that produces fluent, well-formatted, citation-bearing analysis at high speed can lend unearned authority to weak results. The same tooling that industrializes self-refutation can, in less disciplined hands, industrialize the *appearance* of rigor. This is the primary negative societal risk of the methodology.
- **Citation integrity.** AI-assembled bibliographies are prone to plausible-but-wrong references; in this project, a verification pass caught multiple incorrect titles and a misattributed result. Unverified AI citations pollute the scholarly record and must never be published unchecked.
- **Epistemic deskilling.** Habitual reliance on generated controls may atrophy the researcher’s own ability to design them — a long-horizon concern for training the next generation.
- **Equity of access.** Methodologies that presuppose frontier-model access risk widening the gap between resource-rich and resource-poor research groups.
- **Blind spot — shared priors.** The human and the AI may share cultural/training priors (e.g., a background preference for parsimony/ Λ CDM), so their “agreement” is weaker evidence than two genuinely independent analysts. This is treated adversarially in §5.

Reproducibility statement

Reproducibility protocol (placeholders for release)

Code repository: [URL PLACEHOLDER -- e.g. github.com/...]

Analysis scripts: `ab_v3.py` (a_B diagnostic + control), `close_loop2.py` (loop-closure fit), `lrg1_pull.py` (leave-one-out), `ap_basis.py` (AP rotation), `boss_crosscheck.py`, `dr3_forecast.py` (pre-registered test), `bugcheck.py` (controls/nulls).

Datasets: DESY5 Hubble diagram (`des-science/DES-SN5YR`, tag v1.0); DES-Dovekie HD (same repo, main); Pantheon+ (`PantheonPlusSH0ES/DataRelease`); DESI DR2 BAO (arXiv:2503.14738, Table); BOSS DR12 consensus BAO (Alam et al. 2017).

Environment: [PLACEHOLDER: python X.Y; numpy, scipy, pandas, matplotlib versions; OS; seed values]. Null simulations used fixed seeds ([PLACEHOLDER]); MC realizations N = [PLACEHOLDER].

Compute: CPU-only; no GPU or specialized hardware required.

AI assistance disclosure: All AI-generated code was executed and independently bug-checked; all AI-suggested citations were verified against primary sources before inclusion. Model: [PLACEHOLDER: model name / version / date].

Known non-reproducible elements: the compressed-likelihood absolute significances are pipeline-specific; a full-Planck recomputation is required for publication-grade values.

5 Adversarial stress-testing and threat modeling

(Perspective 4: AI red-teaming & vulnerability analyst.)

We evaluate the methodology as an attacker would — asking not “did it work?” but “how could this exact workflow have produced a confident, wrong, well-formatted conclusion, and what in the process would have caught it?” We enumerate four AI-specific failure classes, their manifestation risk here, and the protocol (used or required) that mitigates each.

5.1 Threat 1 — Confirmation-bias feedback loops

Vector. An LLM optimizes for coherence with the conversation. If the human steers toward a systematics/ Λ CDM reading, the model can preferentially generate analyses that confirm it, and the human reads confirmation as independent corroboration — a closed loop with no external check.

Manifestation risk here: high. The investigation *did* trend toward a skeptical conclusion, and the human’s framing was skeptical throughout. This is the single most serious vulnerability.

Mitigations (used). (i) Null simulations with a Λ CDM-*true* generating model, which can only fail the skeptical hypothesis — e.g. the 6.5σ -from-null result would have collapsed had the offset been a fitting artifact. (ii) Sign and dose-response tests: the loop-closure was required to show monotonicity and correct sign, not merely a shift. (iii) The reversals themselves (§2) are evidence the loop was at least partially broken: the model repeatedly produced results that *contradicted* the running narrative (the overclaim retraction, the binning artifact).

Mitigation (required, not fully met). An adversarial second operator instructed to argue the *pro-new-physics* case, or a second model with an opposing prompt, to break the shared-prior blind spot of §4. The real-physics branch was acknowledged as under-tested precisely because of this gap.

5.2 Threat 2 — Hallucination and fabricated specificity

Vector. LLMs generate fluent false specifics: non-existent citations, invented parameter values, plausible-but-wrong equations, or numerical results not actually computed.

Manifestation risk here: confirmed and caught. The AI-assembled bibliography contained multiple incorrect paper titles and one misattributed quantitative result; a dedicated verification pass corrected them against primary sources. Had this pass been skipped, fabricated citations would have entered the record.

Mitigations (used). (i) **Execute, don’t trust:** every numerical claim came from executed code whose output was inspected, not from model assertion. (ii) **Citation verification pass:** each reference checked against arXiv/journal primary sources. (iii) **Re-derivation from scratch** after container resets, which functioned as an inadvertent replication — foundational results (the 7.4σ offset, the $1.6\sigma/3.0\sigma$ leave-one-out) were reproduced independently.

Residual exposure. Execution guards numbers but not their *interpretation*; a hallucinated but plausible physical rationale attached to a correct number is not caught by running the code. This class remains partially open and is the reason interpretive claims are tiered and hedged.

5.3 Threat 3 — Sycophancy / alignment drift

Vector. Over a long session, an assistant may drift toward agreement, softening disagreement and validating the user’s framing to maintain rapport — the analog of alignment drift in extended interaction.

Manifestation risk here: moderate. Detectable counter-evidence: the assistant issued unprompted corrections (retracting its own “removes the bulk” overclaim; flagging the unweighted-binning bug against the running narrative; declining to file “dark energy is vacuum energy” as supported despite the human raising it favorably). These are anti-sycophantic acts.

Mitigations (used / required). (i) Explicit standing instruction to prioritize honesty over agreement and to flag its own errors. (ii) The tiered ledger, which structurally forces every claim to declare its evidence status, making unearned validation visible. (iii) **Required:** periodic “devil’s-advocate” turns in which the assistant is instructed to make the strongest case *against* the current conclusion; used informally, should be formalized.

5.4 Threat 4 — Pipeline integrity and prompt injection

Vector. A research pipeline that ingests web content and executes generated code has an attack surface: malicious instructions embedded in fetched pages (prompt injection), compromised data sources, or generated code with unintended side effects.

Manifestation risk here: low but nonzero. The workflow fetched arbitrary web content (papers, data) and executed generated code with network access.

Mitigations (used / required). (i) Data provenance restricted to known repositories (official DESI/DES/Pantheon+ releases) rather than arbitrary mirrors. (ii) Generated code inspected before execution; no credentialed or destructive operations. (iii) **Required for hardened use:** sandboxed execution with no outbound network during compute; treat all fetched text as untrusted data, never as instructions; checksum/pin dataset versions to defeat source substitution.

5.5 Red-team summary

Threat	Risk here	Primary mitigation (used)	Residual / required
Confirmation loop	High	Λ CDM-true null sims; reversals	Adversarial 2nd operator
Hallucination	Confirmed	Execute-don’t-trust; citation pass	Interpretive claims still hedged
Sycophancy drift	Moderate	Honesty-first instruction; ledger	Formalize devil’s-advocate turns
Pipeline / injection	Low	Trusted sources; code inspection	Sandbox; pin data; distrust fetched text

Table 3: Threat model summary. Two classes (confirmation loop, hallucination) are the material risks; both were partially but not fully mitigated, and the residual exposure is the reason all interpretive conclusions are tiered and provisional.

6 Synthesis and conclusion

The four perspectives converge on one claim. The AI-assisted methodology’s genuine contribution was not answers but *the industrialization of self-refutation*: by collapsing the cost of controls, nulls, and re-derivations, it made trying-to-break-a-result cheaper than believing it, and the investigation’s substantive progress came entirely through the resulting reversals (§2). The comparative analysis (§3) identifies this cost inversion as the transferable lesson for human practice. But the ethics and red-team analyses (§4, §5) show the same property is double-edged: the workflow that cheaply refutes can, undisciplined, cheaply fabricate, and its two material vulnerabilities — the confirmation-bias loop and hallucinated specificity — are intrinsic to LLM collaboration, mitigated here but not eliminated.

The honest verdict on the method mirrors the honest verdict on the physics it studied: *promising, partially validated, and explicitly not settled*. The methodology produced a well-structured, self-corrected, falsifiable body of work, and it did so faster than a lone human could. It did not, and structurally cannot on its own, guarantee the correctness of its interpretive conclusions — which is why its outputs are delivered as a tiered ledger of provisional positions with pre-registered tests, rather than as claims. That framing is not a limitation of the method; it is the method’s most defensible product.

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